VIRTUAL HANDWRITING BASED SMART BOARD USING DEEP LEARNING.

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***Abstract*—Writing is a specific technique for communication that allows us to express our ideas. The two most common methods for recording information are typing and writing. The process involves using a pen or a finger to inscribe letters or words in a relaxed environment. With wearable devices capable of observing and comprehending our actions, gesture recognition has become a computational technique that strives to recognize and interpret human gestures through mathematical algorithms. Our project aims to exploit this technology to develop a motion-to-text converter, a software program for smart wearable devices designed for writing in the air. The application will utilize computer vision to track finger movement, with the resulting text having diverse applications, including text messaging and emailing. This technology will prove beneficial in facilitating communication for people with hearing impairments.**

***Index Terms*—human gesture recognition, motion-to-text converter, computer vision.**

I. Introduction

The artificial intelligence-based virtual smart board is interactive in real time. This project's immediate goal is to improve learning opportunities offered by conventional teaching strategies. Many educational platforms make use of interactive whiteboards. There are numerous whiteboard programs available in the IT industry. These both have benefits and drawbacks. One of the significant limitations of this whiteboard software is the demand for precise pointing devices. Here, By introducing an interactive Virtual Smart Board and a hand motion writing program. Deep Learning models give scribbling functions by recognizing hand gestures. The user can draw without the need for an additional pointing device. Instead, different hand motions made with the fingertips are used as a pen. Text writing, text removal, and color deception are all possible using computer vision technology. Traditional writing software comes with the drawbacks of requiring conventional writing instruments such as pens, a mouse, and other writing tools.

Given the numerous uses for hand gesture recognition in computer gaming, virtual reality, and sign language translation, it is important to realize human-computer interaction (HCI) [1]. Despite much earlier research, traditional vision-based hand motion detection methods still require advancement for practical applications. The ability of optical sensor-based methods to detect and track the hands robustly is typically impossible due to the nature of optical sensing, which is sensitive to lighting conditions and cluttered backgrounds and significantly impacts hand gesture recognition's effectiveness.

Hand gestures are a form of body language [1] characterized by the palm's center, the fingers' placement, and the hand's shape. There are two categories of hand gestures: static and dynamic. The dynamic gesture involves a series of hand gestures, such as waves, whereas the static gesture corresponds to the fixed form of the hand, as its name indicates. Many hand movements make up a gesture; a handshake, for instance, differs from person to person and is location- and time-specific. The key distinction between gesture and posture is that one emphasizes the shape of the hand while the other emphasizes its motion. Wearable glove-based, and camera vision-based sensor techniques are the two basic strategies for analyzing hand gestures.

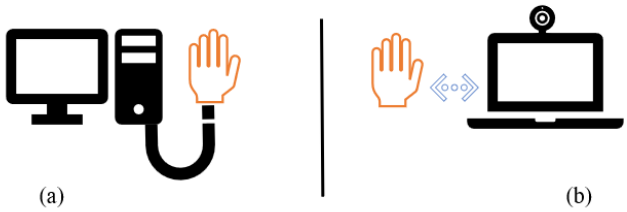

Fig. 1. Illustrates hand gesture approaches. (a) Attached glove-based sensor, connected to a computer; (b) computer vision-based camera, involving merely a bare hand.

Given that they foster interaction and communication and can be used in various contexts, hand gestures serve as a targeted stimulus for study. In the past, hand motions were detected using wearable sensors directly attached to the hands while wearing gloves. These sensors picked up a physical reaction from hand motions and finger bending. After obtaining the data was processed on a computer connected to the glove. The data glove sensor enabled hand gestures in human-computer interaction (HCI)[2], as depicted in Figure 1. It featured simple computer interface commands. By identifying the exact coordinates of the palm and fingers, various sensor types

were used in the gloves to record hand motion and position. Based on the bending angle, the curvature sensor, angular displacement sensor, optical fiber transducer, flex sensor, and accelerometer sensor all employed the same approach [2]. The physical principles that these sensors employ vary depending on the kind.

Writing has evolved significantly over the years. It was invented circa 2000 BC by Neolithic humans who originally wrote on walls before moving on to stones. Cloth eventually supplanted stones; now, paper is the predominant communication medium. Writing, however, has grown progressively digitalized since the advent of the QWERTY keypad. Electronic gadgets are progressively replacing traditional writing materials such as pens and paper.

The ”Finger motion recognition and tracking systems” [3] are a further subdivision of the gesture recognition systems, which is a subtle variant. A finger motion recognition system is perceived to be the most effective way of typing, contributing to the philosophy of ”touchless typing.” Finger tracking systems mainly deal with user-data interaction, intending to make gestures and hand movements quicker and more creative. The primary goal behind the development of finger-tracking systems is to make computers easier to use and control by utilizing natural language and gesture interactions[3].

In this work, we combine this finger motion tracking system with a web camera to provide a much more affordable option, facilitating more straightforward typing. The Summary The methods utilized to develop and put the suggested system into operation are covered in Section 2 of the paper's overview. While Section 4 presents our observations and findings following the system's execution, Section 3 focuses on the implementation approaches. Finally, Section 5 presents the interpretation of the findings.

II. BACKGROUND RELATED WORKS

The quick popularity of augmented and virtual reality boosted the desire to develop human-machine interactions. Hand gesture applications have grown in popularity in recent years, and multiple methods for hand gesture recognition have been developed, including automotive interfaces (Ohn-Bar Trivedi, 2014) [6], cost-effective air writing systems (Pavithra Prabhu, 2016) [4], and handwriting recognition in free space. (Shashidhar, Kim, and Chai, 2015) [5] . However, identifying hand motions alone is insufficient for writing in the air, as it also requires fingertip detection, tracking, and tracing. One way is to use a fingertip-detecting or LED light-detection device developed by Pavithra et al. This approach detects LED lights, records their movement, and then utilizes Optical Character Recognition (OCR) to show the matching alphabet on the screen. However, these device-based techniques have several drawbacks.

Our subsequent hand landmark model employs regression, or direct coordinate prediction, to achieve exact key point localization of 21 3D hand-knuckle coordinates throughout the observed hand areas soon after the palm is recognized over the entire image. [9]. The model can represent reliable internal hand position even with self-occlusions and partially visible hands. As shown in Figure 2, to get ground truth data, we manually inserted 21 3D coordinates to over 30K real-world photos. Additionally, to better cover the range of potential hand positions and offer additional supervision on the nature of the hand, we generate a high-quality synthetic hand geometry model over diverse backgrounds and map it to the relevant 3D coordinates.

Araga et al. created a hand touch identification algorithm that uses Jordan's Recurrent Neural Network (JRNN) to compare representative still photos of 5 or 9 various hand positions and differentiate hand shapes through recording re-installation. After identifying a sequence of positions, the JRNN receives touch input, and a novel training approach was developed, yielding 99.0

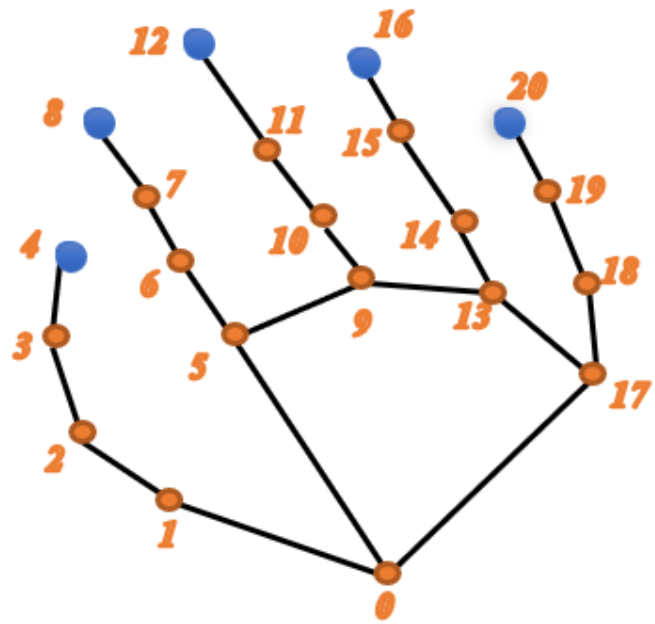

Fig. 2. 0- Wrist, 4- Thumb Fingertip, 8- Index Fingertip, 12- Middle Fingertip, 16- Ring Fingertip, 20- Pinky-Tip

Automatic object tracking has multiple benefits in computer technology, primarily in computer vision and human-machine interaction. Tracking algorithms have been proposed for various applications, including reading sign language signals, detecting hand motions and writing, monitoring body movement, and recognizing characters based on finger tracking. Bragatto et al. created a system for automatically translating Brazilian Sign Language that uses a multilayer neural perceptron network with real-time video capture and processing[11]. Referring to Figure 3, The network was employed in two stages for color identification and hand form validation, with a 99.2

The technology was shown to be appropriate for physical reality applications, and Wang et al. proposed a method for doing so using a webcam and a tracking item attached to a t-shirt to detect both internal and external motion. Jari Hannuksela et al., Toshio Asano et al., and Sharad Vikram

Fig. 3. Brazilian Sign Language Hand Configuration

et al. created finger recognition systems that use finger movement tracking algorithms that combine two Kalman filtering techniques and expected expansion (EM) methods to analyze two different movements, and camera rates are used to count finger movements[12]. Swiping a finger in front of a camera can be used to operate cell phone gadgets. The authors propose visually identifying Japanese katakana characters with an LED pen and camera. Typing speed was tested using traffic codes including up to 100 files, and 46 Japanese symbols, and a single camera achieved 92.9

The current technology can only recognize fingers and cannot distinguish things such as highlighters, paints, or relatives. Identifying and differentiating things such as fingers from RGB images without a depth sensor is difficult. Furthermore, the system has issues with tracking pen movements and placings. Only one RGB camera cannot determine depth, and tracing the pen's up, and down movements become difficult. As a result, the resulting finger route image is abstract and unrecognizable to the system. Real-time hand movements necessitate considerable code changes for seamless transitions between areas. Furthermore, to accurately control their plan, the user must learn a variety of hand movements. The project aims to address some of the most pressing social issues and challenges.

Malathi et al. relate that using face masks as a preventive measure during the Covid-19 pandemic has highlighted the need for efficient and accurate detection of individuals wearing masks to avoid cross-contamination [13]. Manual tracking of individuals wearing masks is costly and time-consuming, leading to the development of automated approaches using Artificial Neural Networks. In particular, the YOLOv3 and MobileNetv2 architectures have shown promise for object detection tasks, including detecting individuals wearing masks. However, further research is needed to determine their performance and accuracy in real-world settings and their suitability for different scenarios, such as varying lighting conditions and facial expressions. This paper proposes a comparison of these architectures to determine their effectiveness for face mask detection, which could improve the accuracy and reliability of mask detection in infection protection strategies.

A weighted average ensemble of low, middle, and high-resolution convolutional neural networks (CNNs) [14] with enhanced versions of VGG-16 is used in the proposed SignNet model by M. Geetha et al. for the recognition of sign language. The ensemble's performance was evaluated using a synthetic dataset, achieving an impressive accuracy of over 92

Aswathi Premkumar et al., in their paper, exhibit that the development of a reliable sign language recognition system [15] is critical to bridging communication barriers between deaf individuals and the hearing majority. Deep learning and computer vision techniques have recently shown promise in accurate sign language recognition. This study suggests the use of the convolutional neural network (CNN) and VGG-16 as two deep learning models for recognizing and categorizing hand motions in American Sign Language. The study optimizes accuracy by analyzing the models' performance using a limited dataset. Both models performed remarkably well, with VGG-16 achieving higher accuracy of 99.56

C. Kavinkumar et al. describe the importance of hand hygiene in preventing the spread of diseases has been recognized by the World Health Organization (WHO). Only a few people adhere to the WHO's suggested seven steps for proper handwashing hygiene. With the COVID-19 pandemic, hand hygiene has become more critical, particularly for healthcare workers at greater risk of exposure to infectious diseases. Although technologies like UV images are available to check hand hygiene, there is no real-time system to monitor the efficiency of a person's handwashing actions. This paper[16] proposes a deep learning model with transfer learning (VGG16) to predict the accuracy of human handwashing actions. This technology can be used in public places and hospitals to monitor hand hygiene and reduce the risk of spreading communicable diseases.

III. PROPOSED METHODOLOGY

The proposed system aims to eliminate the need to use cell phones to take notes. It relies on fingertip detection and finger movement techniques to function effectively. The system initially detects the fingertip using Python, OpenCV, and CNN techniques before tracing its trajectory and displaying it on the screen.

The goal is to enable freehand drawing in an open space. An RGB camera detects the fingertip and tracks its movement across the screen. Finding the fingertip is the first thing to do when the hand is before the camera, and there are different ways to achieve this, refer to Figure 4.

A. Hand Detection training with Yolo v2

Data Collection: The first step is to collect a large dataset of images that contain hands in different poses, orientations, and lighting conditions. The dataset should include various hand shapes, sizes, and skin tones to ensure the algorithm can recognize hands from different people.

Data Annotation: The next step is to annotate the dataset with bounding boxes around each hand in the images. It involves manually drawing a box around each hand and

labeling it with a class name (e.g., ”hand”). This annotated dataset will train the YOLO v2 algorithm to recognize and localize hands in new images.

Data Pre-processing: Before training the YOLO v2 algorithm, the dataset needs to be pre-processed to prepare the images for input into the network. This involves resizing the images to a fixed size, converting them to a standardized format (e.g., RGB), and normalizing the pixel values.

Model Configuration: The YOLO v2 algorithm needs to be configured with the appropriate hyperparameters, such as the number of classes, the number of anchor boxes, and the learning rate. The size of the fully connected layers as well as the number and size of the convolutional layers may be adjusted to modify the model design further.

Training: The next step is to train the YOLO v2 algorithm using the annotated dataset. This involves running the dataset through the network multiple times and adjusting the weights and biases of the neurons to minimize the loss function. The training process can take several hours or days, depending on the dataset’s size and the model’s complexity.

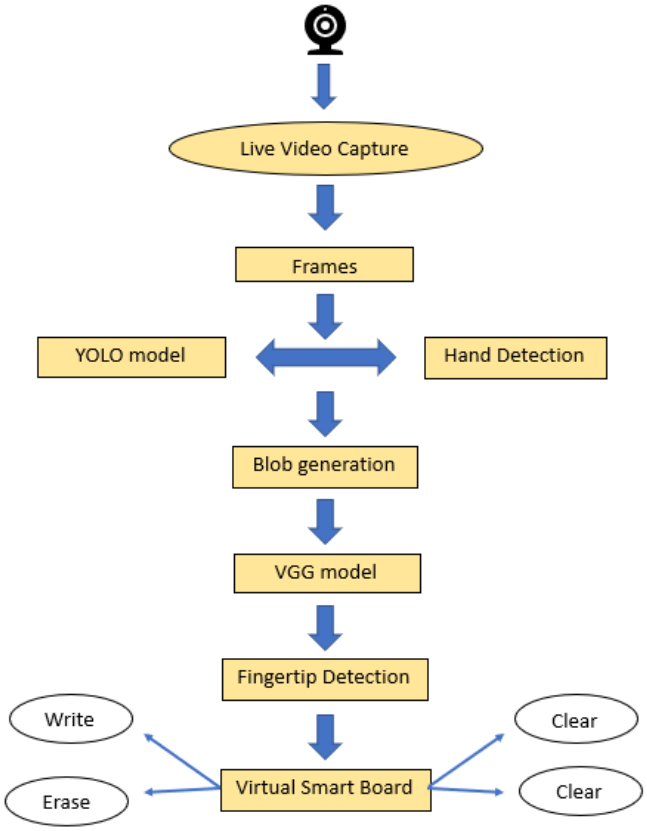

Fig. 4. Proposed Algorithm workflow

Evaluation: For evaluating the YOLO v2 algorithm’s performance after the training, a different validation dataset must be used. This involves calculating metrics to assess how well the system can detect and recognize hands in new images, including precision, recall, and F1 score. Testing: Finally, the YOLO v2 algorithm can be tested on a different test dataset to evaluate its effectiveness in real-world scenarios. This involves running the algorithm on new images and analyzing its output to see how well it can detect and localize hands; partial results are attached in Figure 5.

B. Fingertip detection

We aim to create a fingertip detection system that can provide accurate results. Initially, we detect the entire hand, followed by region segmentation. The region segmentation process involves two steps, namely skin segmentation and background subtraction. This system is designed to function in real-time and is expected to deliver precise outcomes. We may employ faster RCNN methods for background subtraction.

The calculation of the center of gravity holds significant importance as it aids in detecting specific hand gestures for performing operations. The proposed system utilizes two algorithms to calculate the centroid and, subsequently, takes the average value of both algorithms to obtain the final result. The algorithm employed is distance transformation, and the pixel with the highest intensity is determined as the center of gravity.

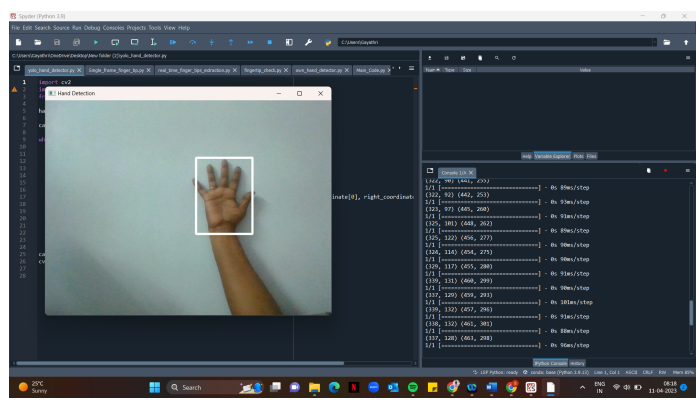

Fig. 5. Hand detection in the real-time video through a basic webcam

C. Fingertip training with VGG net

Data collection: Collect a dataset of images containing hands with visible fingers and tips. The dataset should contain enough variation in hand poses, lighting conditions, and backgrounds to make the network robust to different conditions.

Data pre-processing: Pre-process the images by scaling them to a fixed size, pixel-value normalization, and dataset augmentation with random transformations such as rotation, translation, and flipping.

Training the VGG net: Train a VGG network on the pre-processed dataset using supervised learning. The network should be trained to predict the coordinates of the fingertips in the image.

Validation: Validate the trained network on a separate validation set to ensure that it can generalize well to new images.

Testing: Test the network on a test set to evaluate its performance on fingertip detection.

Post-processing: After obtaining the predicted coordinates of the fingertips, post-process the coordinates to filter out false detections and refine the coordinates using techniques such as non-maximum suppression and regression, partial results are attached in Figure 6.

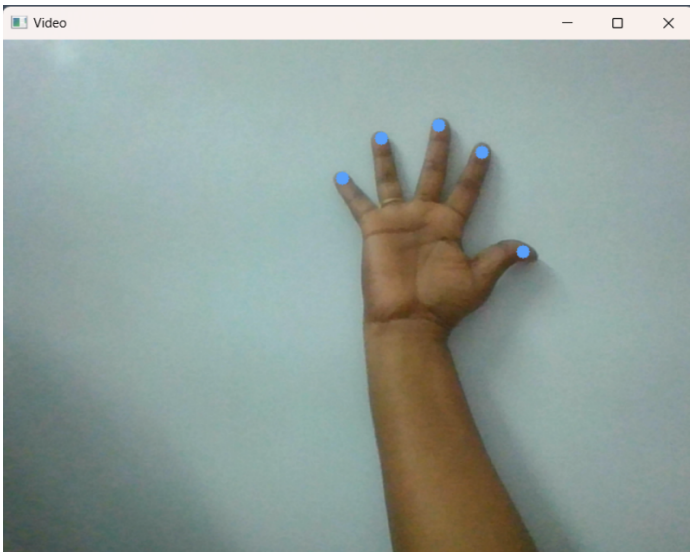

Fig. 6. Fingertip Recognition in real-time video by webcam.

D. Fingertip Tracking

Once the hand region and center of gravity have been successfully detected, the next step is to track the movement of the fingertips on the screen. Previous research has shown that the faster R-CNN handheld detector is resource-intensive, resulting in frame rates below real-time performance. Therefore, our goal is to use the KCF tracking algorithm. This algorithm converts the detected fingertip into the HSV color space refer to Figure 7 8.

Once the mask is identified in the air, the system executes morphological operations to eliminate impurities from the masked image. Following the detection of contours, drawing the line becomes the most crucial step. To achieve this, a python deque is established. The deque retains the outline's position in each consecutive frame, which creates a line utilizing OpenCV's drawing capabilities.

IV. Experimental Analysis

Detection of Fingertips: Experimental Results We used the EgoFinger dataset, which consists of 12,974 images and was previously used in [14,15], to evaluate the effectiveness of the proposed method against those of existing techniques. Our evaluation metric, the Average Pixel Error (APE), which calculates how far apart the identified position is from the actual place, demonstrates that the approach we propose has a substantially less APE., indicating greater accuracy in locating the fingertip. In Figure 8, you can see that we also plotted the percentages of successful detection at different APE threshold settings. A successful detection is one in which the pixel-based

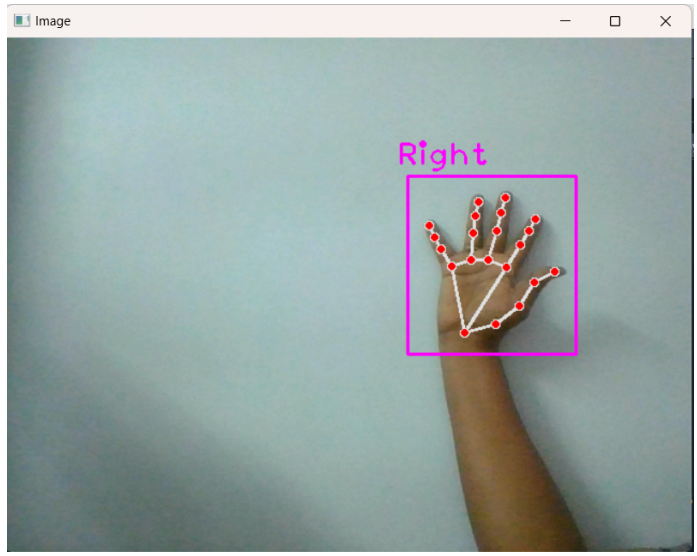

Fig. 7. Fingertip tracking with own hand detection

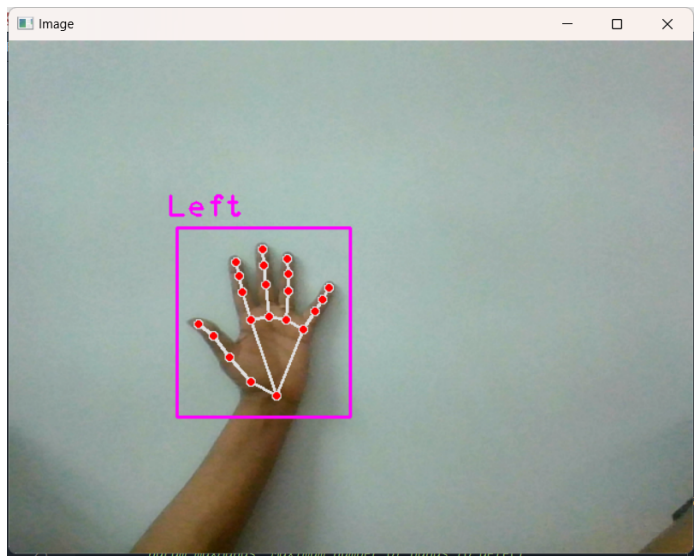

Fig. 8. Fingertip tracking with own hand detection

threshold is less than when compared to the detected fingertip's location concerning the ground truth. When implemented, our strategy had a more than 90

Our proposed scheme was compared to four other algorithms, including FRCNN [16], KCF [29], TLD [30], and MIL [31]. We chose a threshold of 15 pixels to assess how well the detection worked. Due to the backbone's unique design, we could still use the EgoFinger test dataset in our experiments, even though the test dataset was absent from these studies. As illustrated, our proposed scheme can accurately detect fingertip locations across diverse scenes.

A. Project Scope

The primary application of this system is to serve as an efficient mode of communication for individuals who are deaf.

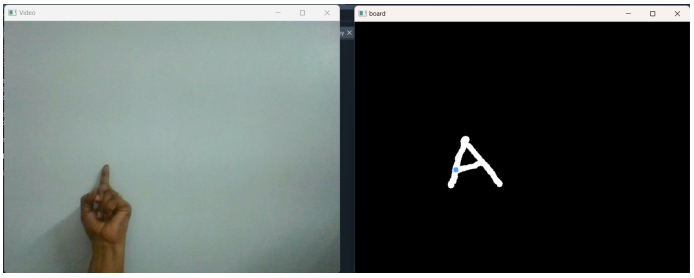

Fig. 9. In Real Time, Fingertip detected and writing function enabled on board as shown in the above picture.

By implementing this project, we can achieve the following benefits:

1) Removing the necessity for writing offers a powerful communication tool that lessens dependency on cell phones and PCs.

2) People with hearing impairments can communicate more effectively thanks to it.

3) Various uses, including sending emails and messages, are possible for the created text.

CONCLUSION

This program has the potential to revolutionize traditional note-taking techniques by doing away with the need for a mobile phone in the hand and offering a convenient on-the-go option. It can significantly assist individuals with hearing impairments and those who struggle with keyboard usage. Furthermore, as the program's functionality expands, it may be used to control IoT devices and even enable air painting. The program could also be helpful in innovative clothing software, facilitating improved digital world interaction. Through this system, text can come alive, and advanced discovery algorithms like YOLO v3 can enhance fingerprint recognition accuracy and speed. With continued advancements in artificial intelligence, we can expect even greater efficiency in air writing.

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